

KENDRIYA VIDYALAYA SANGATHAN (PATNA REGION)

CLASS: XII

FIRST PRE-BOARD EXAMINATION

SESSION – 2024 – 25

SUBJECT: PHYSICS (THEORY)

Maximum Marks: 70

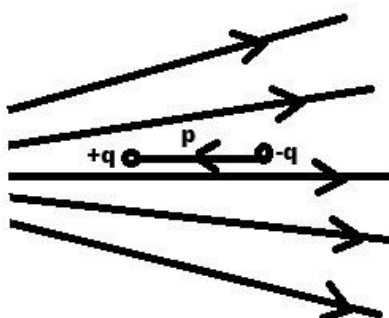
Time: 03 Hours

General Instructions:

- (1) There are 33 questions in all. All questions are compulsory.
- (2) This question paper has five sections: Section A, B, C, D and E.
- (3) In section – A: question number 1 to 16 are Multiple Choice (MCQ) type questions carrying 1 mark each.
- (4) In section – B: question number 17 to 21 are Short Answer-1 (SA-1) type questions carrying 2 marks each.
- (5) In section – C: question number 22 to 28 are Short Answer-2 (SA-2) type questions carrying 3 marks each.
- (6) In section – D: question number 29 to 30 are case-based questions carrying 4 marks each.
- (7) In section – E: question number 31 to 33 are Long Answer (LA) type questions carrying 5 marks each.
- (8) There is no overall choice. However, an internal choice has been provided in 2 questions in Section – B, 2 questions in Section – C, 2 questions in Section – D and 3 questions in Section – E.

SECTION – A

- (1.) Figure shows electric field lines in which an electric dipole p is placed as shown. Which of the following statement is correct?



- (a) the dipole will not experience any force.
- (b) the dipole will experience a force towards right.
- (c) the dipole will experience a force towards left.
- (d) the dipole will experience a force upwards.

(2) Which of the following is non-ohmic resistance?

- (a) Lamp filament (b) Copper wire
(c) Carbon resistor (d) Diode

(3.) A conducting wire of 1 m length is used to form a circular loop. If it carries a current of 1 ampere, its magnetic moment will be _____ Am².

- (a) 2π (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) $\frac{1}{4\pi}$

(4.) A bar magnet AB with magnetic moment M is cut into two equal parts perpendicular to its axis. One part is kept over the other so that end B is exactly over A. What will be the magnetic moment of the combination?

- (a) Zero (b) $\frac{M}{4}$ (c) $\frac{3M}{4}$ (d) M

(5.) Two coils are placed close to each other. The mutual inductance of the pair of coils depends upon the

- (a) rate at which current change in the two coils.
(b) relative position and orientation of the coils.
(c) rate at which voltage induced across two coils.
(d) Current in the two coils.

(6.) In a step-down transformer, the number of turns in the secondary coil is 20 and the number of turns in the primary coil is 100. If the voltage applied to the primary coil is 120 V, then what is the voltage output from the secondary coil?

- (a) 24 V (b) 12 V (c) 6 V (d) 18 V

(7.) What is the phase difference between electric and magnetic fields in an electromagnetic wave?

- (a) 0 (b) π (c) $\pi/2$ (d) $\pi/4$

(8.) How does the magnifying power of a telescope change on increasing the linear diameter of its objective?

- (a) Power increases on increases diameter
(b) Power decreases on decreases diameter
(c) Power remains constant on increases diameter
(d) Power doesn't depend on diameter

(9.) In a single-slit diffraction experiment, the width of the slit is halved. The width of the Central maximum, in the diffraction pattern, will become

- (a) one-fourth (b) half (c) twice (d) four time

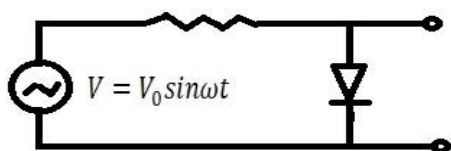
(10.) The photoelectric effect is based on the law of conservation of

- (a) mass (b) momentum
(c) energy (d) none of these

(11.) When an electron in an atom goes from a lower to a higher orbit, its

- (a) kinetic energy decreases, potential energy increases
- (b) kinetic energy increases, potential energy decreases
- (c) kinetic energy increases, potential energy increases
- (d) kinetic energy decreases, potential energy decreases

(12.) The output of the given circuit shown in figure



- (a) would be zero at all times.
- (b) Would be like a half wave rectifier with negative cycles in output.
- (c) Would be like a half wave rectifier with positive cycles in output.
- (d) Would be like that of a full wave rectifier.

Note : In question number 13 – 16 two statements are given – one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below :

- (a) Both Assertion (A) and Reason (R) are true and (R) is the correct explanation of (A).
 - (b) Both Assertion (A) and Reason (R) are true and (R) is NOT the correct explanation of (A).
 - (c) Assertion (A) is true and Reason (R) is false.
 - (d) Assertion (A) is false and Reason (R) is true.
- (13.) Assertion : The total charge stored in a capacitor is zero.
Reason : The field just outside the capacitor is σ/ϵ_0 . (σ is the charge density).
- (14.) Assertion : The image of an extended object placed perpendicular to the principal axis of a mirror, will be erect if the object is real but the image is virtual.
Reason : The image of an extended object, placed perpendicular to the principal axis of a mirror, will be erect if the object is virtual but the image is real.

(15.) Assertion : Bohr had to postulate that the electrons in stationary orbits around the nucleus do not radiate.
Reason: According to classical physics all moving electrons radiate.

(16.) Assertion : A photon has no rest mass, yet it carries some momentum.
Reason : Momentum of photon is due to its energy and due to its equivalent mass.

SECTION – B

(17.) A cell of emf E and internal resistance r is connected across a variable resistor R . Plot a graph showing variation of terminal voltage V of the cell versus the current I . Using the Plot, show how the emf of the cell and its internal resistance can be determined.

(18.) A closely wound solenoid 80 cm long has 5 layers of winding of 400 turns each. The diameter of the solenoid is 1.8 cm. If the current carried is 8.0 A, estimate the magnitude of $\vec{B}$ inside the solenoid near its centre.

(19.) A concave lens of refractive index 1.5 is immersed in a medium of refractive index 1.65. What is the nature of the lens?

OR

What is the focal length of a combination of a convex lens of focal length 30 cm and a concave lens of focal length 20 cm in contact? Is the system a converging or a diverging lens? Ignore thickness of lenses.

(20.) Plot suitable graphs to show the variation of photoelectric current with the collector plate potential for the incident radiation of

(i) the same intensity but different frequencies ν_1, ν_2 and ν_3 ($\nu_1 < \nu_2 < \nu_3$)

(ii) the same frequency but different intensities I_1, I_2 and I_3 ($I_1 < I_2 < I_3$)

OR

Plot a graph showing variation of de-Broglie wavelength λ versus $\frac{1}{\sqrt{V}}$, where V is accelerating potential for two particles A and B carrying same charge but of masses m_1, m_2 ($m_1 > m_2$). Which one of the two represents a particle of smaller mass and why?

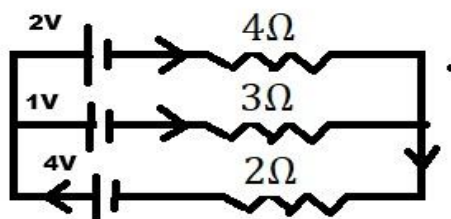
(21.) Draw energy band diagram of n-type and p-type semi-Conductors at temperature $T > 0K$, depicting the donor and acceptor energy levels. Mention the significance of these levels.

SECTION – C

- (22.) (a) Draw a graph of electric field E versus a distance r from the centre of the electric dipole for $r \gg a$.
- (b) If an electric dipole were kept in a uniform external electric field E , diagrammatically represent the position of the dipole in stable and unstable equilibrium and write the expression for the torque acting on the dipole in both cases.
- (23.) In a Wheatstone bridge, $P = 1 \Omega$, $Q = 23 \Omega$, $R = 2 \Omega$, $S = 3 \Omega$ and the galvanometer resistance is 4Ω . Find the current through the galvanometer in the unbalanced position of the bridge, when a battery of 2 V and internal resistance 2Ω is used.

OR

State Kirchhoff's Laws. Use these laws, calculate the currents I_1 , I_2 and I_3 in the following circuit diagram:



- (24.) Write three points of differences between dia-, para- and ferro-magnetic materials, giving one example of each.
- (25.) An ac source of voltage $V = V_0 \sin \omega t$ is connected to a series combination of L , C and R . Use the phasor diagram to obtain expression for impedance of the circuit and phase angle between voltage and current.
- (26.) (a) Sketch a schematic diagram depicting oscillating electric and magnetic fields of an em wave propagating along $+x$ -direction.
- (b) Which of the following electromagnetic waves has minimum wavelength and minimum frequency? Write one use of each of these two waves.

Infrared waves, Microwaves, λ -rays and X-rays

- (27.) Determine the distance of closest approach when an alpha particles of kinetic energy 4.5 MeV strikes a nucleus of $Z = 80$, stops and reverses its direction.

OR

An electron jumps from fourth to first orbit in an atom. How many number of spectral lines can be emitted by the atom? To which series these lines correspond?

- (28.) A given coin has a mass of 3.0 g. Calculate the nuclear energy that would be required to separate all the neutrons and protons from each other. For simplicity assume that the coin is entirely made of $^{63}\text{Cu}_{29}$ atoms (of mass 62.92960 u). The masses of proton and neutrons are 1.00783 u and 1.00867 u respectively.**

SECTION – D

Note : Questions number 29 and 30 are Case Study based questions.

Read the following paragraph and answer the questions.

- (29.) Read the following paragraph and answer the questions that follow:**

There are different techniques of fabrication of p-n junction. In one such technique, called fused junction techniques, an aluminium film is kept on the wafer of n-type semiconductor and the combination is then heated to a high temperature (about 600°C). As a result, aluminium fused into silicon and produces p-type semiconductor and in this way a p-n junction is formed.

- (i) When a p-n junction is reversed biased, then how does the height of potential barrier change?**

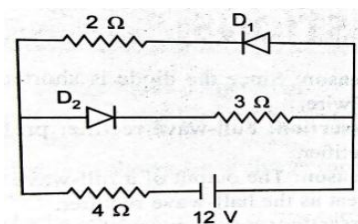
- (a) no current flows**
- (b) the depletion region is reduced**
- (c) height of potential barrier is decreased**
- (d) height of potential barrier increased**

- (ii) The cause of potential barrier in p-n junction is**

- (a) depletion of positive charges near the junction.**
- (b) depletion of negative charges near the junction.**
- (c) concentration of negative charges near the junction.**
- (d) concentration of positive and negative charges near the junction.**

- (iii) The circuit has two oppositely connected ideal diodes in parallel.**

What is the current flowing in the circuit?



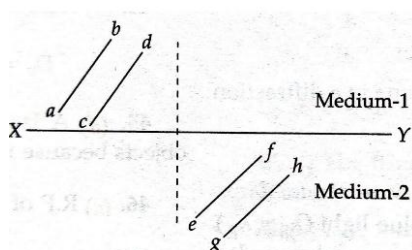
- (a) 1.17 A (b) 1.33 A (c) 2.0 A (d) 2.31 A**

- (iv) When a p-n junction is forward biased, then
- (a) only diffusion current flows.
 - (b) only drift current flows
 - (c) both diffusion current and drift current flow but drift current exceeds the diffusion current.
 - (d) both diffusion current and drift current flow but diffusion current is more than drift current.

OR

- (v) Carbon, germanium and silicon all are fourteenth group element.
- (a) C and Ge are semiconductor.
 - (b) C and Ge are semiconductor.
 - (c) Si and Ge are semiconductor.
 - (d) all C, Si and Ge are semiconductor.

(30.) Wavefront is a locus of points which vibrates in same phase. A ray of light is perpendicular to the wavefront. According to Huygens principle, each point of the wavefront is the source of a secondary disturbance and the wavelets connecting from these points spread out in all directions with the speed of wave. The figure shows a surface XY separating two transparent media, medium- 1 and medium-2. The lines ab and cd represent wavefronts of a light wave travelling in medium- 1 and incident on XY. The lines ef and gh represent wavefronts of the light wave in medium -2 after refraction.



- (i) Select the right option in the following:
- (a) Christian Huygen's a contemporary of Newton's established the wave theory of light by assuming that light waves were transverse.
 - (b) Maxwell provided the theoretical evidence that light is transverse wave.
 - (c) Thomas Young experimentally proved the wave behaviour of light and Huygen's assumption.
 - (d) All the statements given above correctly answers the question "What is light"?

(ii) Wavefront is the locus of all points, where the particles of the medium vibrate with the same

(a) phase

(b) amplitude

(c) frequency

(d) period

(iii) Speed of light is

(a) same in both medium.

(b) larger in medium – 1 than medium -2.

(c) larger in medium – 2 than medium -1.

(d) different at b and d.

(iv) A point source that emits waves uniformly in all directions, produces wavefronts that are

(a) spherical

(b) elliptical

(c) cylindrical

(d) plane

OR

(v) What are the types of wavefronts?

(a) Spherical

(b) Cylindrical

(c) Plane

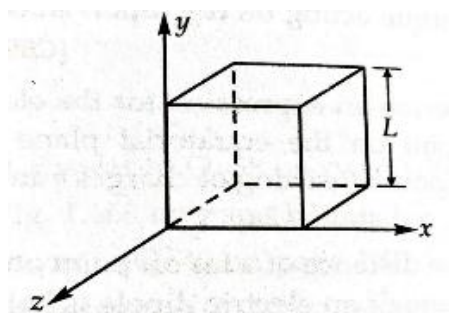
(d) All of these

SECTION - E

(31.) (a) Define electric flux and write its SI unit.

(b) Use Gauss' law to obtain the expression for the electric field due to a uniformly charged infinite plane sheet.

(c) A cube of side L is kept in space as shown in adjoining figure



An electric field $\vec{E} = (Ax + B)\hat{i}$ N/C exists in the region. Find the net charge enclosed by the cube.

OR

- (a) Find out capacitance of a capacitor with a dielectric slab of thickness t between the capacitor plates having separation d .
- (b) What happens when the dielectric slab is replaced by a conducting slab of same thickness?
- (c) How would you connect two capacitors across a battery in series or parallel, so that they store greater (i) total charge and (ii) total energy.

(32.) (a) With the help of a diagram, explain the principle and working theory of a moving coil galvanometer. Justify the necessity of using radial magnetic field in it.

- (b) In a galvanometer, there is a deflection of 10 divisions per mA. The internal resistance of the galvanometer is $78\ \Omega$. If a shunt of $2\ \Omega$ is connected to the galvanometer and there are 75 divisions in all on the scale of the galvanometer, calculate the maximum current which the galvanometer can read.

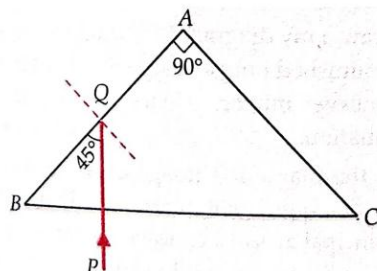
OR

- (a) State Biot-Savart's law. Apply this law to derive an expression for the magnetic field at the centre of a current-carrying circular loop.

- (b) Two identical circular coils, P and Q each of radius R , carrying currents 1 A and $\sqrt{3}\text{ A}$ respectively, are placed concentrically and perpendicular to each other lying in the XY and YZ planes. Find the magnitude and direction of the net magnetic field at the centre of the coils.

(33.) (a) Discuss the variation of the angle of deviation with that of the angle of incidence for a ray of light passing through a prism. Derive an expression for the refractive index of the material of a prism in terms of the angle of prism and the angle of minimum deviation.

- (b) A ray of light PQ enters an isosceles right-angled prism ABC of refractive index 1.5 as shown in adjoining figure:



- (i) Trace the path of the ray through the prism.**
- (ii) What will be the effect on the path of the ray if the refractive index of the prism is 1.4?**

OR

(a) Draw a ray diagram for final image formed at least distance of distinct vision (D) by a compound microscope and write an expression for its magnifying power.

(b) An angular magnification of 22 X is desired for a compound microscope using an objective of focal length 1.5 cm and eyepiece of focal length 2.5 cm. How will you set up the compound microscope (for the final image formed at the least distance of distinct vision)?
